

HDOC – Experiment 7

Question 1

Write a C program to find the total volume and total surface area of a solid made by joining a cone, cylinder, and hemisphere with a common radius.

❖ Understanding problem

Input

- Radius of cone, cylinder and hemisphere (r)
- Height of cone (h_1)
- Height of cylinder (h_2)

Output

- Total Volume
- Total Surface Area

Formula

Volume of Cone:

$$V_c = 1/3(\pi r^2 h_1)$$

Volume of Cylinder:

$$V_{cy} = \pi r^2 h_2$$

Volume of Hemisphere:

$$V_h = 2/3(\pi r^3)$$

Total Volume:

$$V = V_c + V_{cy} + V_h$$

Slant Height of Cone:

$$l = \sqrt{r^2 + h_1^2}$$

Curved Surface Area of Cone:

$$CSA_c = \pi r l$$

Curved Surface Area of Cylinder:

$$CSA_{cy} = 2\pi r h_2$$

Curved Surface Area of Hemisphere:

$$CSA_h = 2\pi r^2$$

Total Surface Area:

$$TSA = \pi r l + 2\pi r h_2 + 2\pi r^2$$

❖ Test case table

Test Case	Input(r,h ₁ ,h ₂)	Expected output	Actual output	Result
1	3,4,5	Volume = 212.058 m ³ TSA = 188.496 m ²	Same	Pass
2	2,5,4	Volume = 87.965 m ³ TSA = 103.003 m ²	Same	Pass
3	5,6,8	Volume = 863.938 m ³ TSA = 502.655 m ²	Same	Pass

❖ Algorithm

1. Start.
2. Read radius, cone height and cylinder height.
3. Calculate slant height.
4. Calculate cone volume.
5. Calculate cylinder volume.
6. Calculate hemisphere volume.
7. Add all three volumes.
8. Calculate cone curved surface area.
9. Calculate cylinder curved surface area.
10. Calculate hemisphere curved surface area.
11. Add all curved surface areas.
12. Display total volume and total surface area.
13. Stop.

❖ Flowchart

Start



Input radius, cone height, cylinder height



Calculate slant height



Calculate cone volume



Calculate cylinder volume



Calculate hemisphere volume



Add volumes

↓

Calculate cone CSA

↓

Calculate cylinder CSA

↓

Calculate hemisphere CSA

↓

Add surface areas

↓

Display Total Volume and TSA

↓

Stop

❖ Program code in C

```
#include <stdio.h>
```

```
#include <math.h>
```

```
#define PI 3.141592653589793
```

```
int main()
```

```
{
```

```
    double r, hCone, hCylinder;
```

```
    double slantHeight;
```

```
    double volume, surfaceArea;
```

```
    printf("Enter common radius: ");
```

```
    scanf("%lf", &r);
```

```
printf("Enter cone height: ");
```

```
scanf("%lf", &hCone);
```

```
printf("Enter cylinder height: ");
```

```
scanf("%lf", &hCylinder);
```

```
slantHeight = sqrt((r * r) + (hCone * hCone));
```

```
volume = (PI * r * r * hCone) / 3.0
```

```
    + (PI * r * r * hCylinder)
```

```
    + (2.0 * PI * r * r * r) / 3.0;
```

```
surfaceArea = (PI * r * slantHeight)
```

```
    + (2 * PI * r * hCylinder)
```

```
    + (2 * PI * r * r);
```

```
printf("\nTotal Volume = %.3lf cubic units\n", volume);
```

```
printf("Total Surface Area = %.3lf square units\n", surfaceArea);
```

```
return 0;
```

```
}
```

❖ Compile, Execute and Evaluate

Test case 1

```
(kali㉿kali)-[~/Desktop/exp7]
$ ./solid
Enter common radius: 3
Enter cone height: 4
Enter cylinder height: 5

Total Volume = 235.619 cubic units
Total Surface Area = 197.920 square units
```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp7]
$ ./solid
Enter common radius: 2
Enter cone height: 5
Enter cylinder height: 4

Total Volume = 87.965 cubic units
Total Surface Area = 109.234 square units
```

Test case 3

```
(kali㉿kali)-[~/Desktop/exp7]
$ ./solid
Enter common radius: 5
Enter cone height: 6
Enter cylinder height: 8

Total Volume = 1047.198 cubic units
Total Surface Area = 531.090 square units
```

EVALUATION: All three test cases produced the expected values. Hence, the program is correct.

❖ Compilation and Execution screenshot

```
(kali㉿kali)-[~/Desktop/exp7]
$ vi solid.c

(kali㉿kali)-[~/Desktop/exp7]
$ gcc solid.c -o solid -lm

(kali㉿kali)-[~/Desktop/exp7]
$ ./solid
Enter common radius: 3
Enter cone height: 4
Enter cylinder height: 5

Total Volume = 235.619 cubic units
Total Surface Area = 197.920 square units

(kali㉿kali)-[~/Desktop/exp7]
$ ./solid
Enter common radius: 2
Enter cone height: 5
Enter cylinder height: 4

Total Volume = 87.965 cubic units
Total Surface Area = 109.234 square units

(kali㉿kali)-[~/Desktop/exp7]
$ ./solid
Enter common radius: 5
Enter cone height: 6
Enter cylinder height: 8

Total Volume = 1047.198 cubic units
Total Surface Area = 531.090 square units
```

Question 2

Write a C program to find the cost of painting an annulus-shaped board when outer radius, inner radius and painting rate per square centimeter are given.

❖ Understanding problem

Input

- Outer radius (R) in cm
- Inner radius (r) in cm
- Painting rate (₹/cm²)

Output

- Area of Annulus
- Painting Cost

Formula

Area of Annulus:

$$A = \pi(R^2 - r^2)$$

Painting Cost:

$$\text{Cost} = A \times \text{Rate}$$

❖ Test case table

Test Case	Input (R,r,Rate)	Expected output	Actual output	Result
1	10,6,2	Area = 201.062 cm ² Cost = ₹402.124	Same	Pass
2	15,10,5	Area = 392.699 cm ² Cost = ₹1963.495	Same	Pass
3	20,12,3	Area = 703.717 cm ² Cost = ₹2111.151	Same	Pass

❖ Algorithm

1. Start.
2. Read outer radius.
3. Read inner radius.
4. Read painting rate.
5. Calculate annulus area.
6. Calculate painting cost.
7. Display area and cost.
8. Stop.

❖ Flowchart

Start

↓

Input outer radius

↓
Input inner radius
↓
Input painting rate
↓
Calculate annulus area
↓
Calculate painting cost
↓
Display area and cost
↓
Stop

❖ Program code in C

```
#include <stdio.h>

#define PI 3.141592653589793

int main()
{
    double outerRadius, innerRadius;
    double rate;
    double area, cost;

    printf("Enter outer radius (cm): ");
    scanf("%lf", &outerRadius);

    printf("Enter inner radius (cm): ");
```

```
scanf("%lf", &innerRadius);
```

```
printf("Enter painting rate (per sq cm): ");
```

```
scanf("%lf", &rate);
```

```
if (outerRadius <= innerRadius)
```

```
{
```

```
    printf("Outer radius must be greater than inner radius.\n");
```

```
    return 1;
```

```
}
```

```
area = PI * ((outerRadius * outerRadius)
```

```
    - (innerRadius * innerRadius));
```

```
cost = area * rate;
```

```
printf("\nArea of Annulus = %.3lf sq.cm\n", area);
```

```
printf("Painting Cost = Rs. %.3lf\n", cost);
```

```
return 0;
```

```
}
```

❖ Compile, Execute and Evaluate

Test case 1

```
(kali㉿kali)-[~/Desktop/exp7]  
$ ./annulus  
Enter outer radius (cm): 10  
Enter inner radius (cm): 6  
Enter painting rate (per sq cm): 2  
  
Area of Annulus = 201.062 sq.cm  
Painting Cost = Rs. 402.124
```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp7]  
$ ./annulus  
Enter outer radius (cm): 15  
Enter inner radius (cm): 10  
Enter painting rate (per sq cm): 5  
  
Area of Annulus = 392.699 sq.cm  
Painting Cost = Rs. 1963.495
```

Test case 3

```
(kali㉿kali)-[~/Desktop/exp7]  
$ ./annulus  
Enter outer radius (cm): 20  
Enter inner radius (cm): 12  
Enter painting rate (per sq cm): 3  
  
Area of Annulus = 804.248 sq.cm  
Painting Cost = Rs. 2412.743
```

EVALUATION: The program successfully computed the annulus area and painting cost for all test cases. The outputs matched the expected values.

❖ Compilation and Execution screenshot

```
(kali㉿kali)-[~/Desktop/exp7]
$ vi annulus.c

(kali㉿kali)-[~/Desktop/exp7]
$ gcc annulus.c -o annulus

(kali㉿kali)-[~/Desktop/exp7]
$ ./annulus
Enter outer radius (cm): 10
Enter inner radius (cm): 6
Enter painting rate (per sq cm): 2

Area of Annulus = 201.062 sq.cm
Painting Cost = Rs. 402.124

(kali㉿kali)-[~/Desktop/exp7]
$ ./annulus
Enter outer radius (cm): 15
Enter inner radius (cm): 10
Enter painting rate (per sq cm): 5

Area of Annulus = 392.699 sq.cm
Painting Cost = Rs. 1963.495

(kali㉿kali)-[~/Desktop/exp7]
$ ./annulus
Enter outer radius (cm): 20
Enter inner radius (cm): 12
Enter painting rate (per sq cm): 3

Area of Annulus = 804.248 sq.cm
Painting Cost = Rs. 2412.743
```